

Integral #1 (Hard)

$$I = \int_0^{\pi/2} \ln(\csc \theta) \sin^3 \theta \, d\theta$$

Solution Methodology:

- Sub: $x = \cos \theta$
- Simplify
- Sub: $u = 1 + x$
- Simplify
- IBP ($\times 2$)

$$I = \frac{5}{9} - \frac{2}{3} \ln 2$$

Integral #2 (Hard)

$$I = \int_0^1 e^{-t^2} f(t) \, dt \quad \text{with} \quad f(t) = \sum_{n=0}^{\infty} e^{-2n(t+n/2)}, \quad t \in \mathbb{R}^+$$

Solution Methodology:

- Complete the square: $-2n(t + n/2) = t^2 - (t + n)^2$
- Simplify Integrand
- Swap summation & integration operators: $\int_0^1 \sum_{n=0}^{\infty} = \sum_{n=0}^{\infty} \int_0^1$
- Sub: $x = t + n$
- Sum the integrals: $\sum_{n=0}^{\infty} \int_n^{n+1} = \int_0^{\infty}$
- Abuse even symmetry: $\int_0^{\infty} = \frac{1}{2} \int_{-\infty}^{\infty}$
- Identify the Gaussian Integral

$$I = \frac{\sqrt{\pi}}{2}$$

Integral #3 (Easy)

$$I = \int_0^{\infty} (2x)^{1012.5} e^{-\sqrt{2x}} \, dx$$

Solution Methodology:

- Sub: $u = 2x, t = \sqrt{u}$
- Identify Gamma Function (or do IBP $\times 2026$)

$$I = 2026!$$

Integral #4 (Medium)

$$I = \int \ln(x + \sqrt{x^2 - 1}) dx$$

Solution Methodology:

- Sub: $x = \cosh \varphi$
- Simplify
- IBP
- Reverse Sub: $\varphi = \cosh^{-1} x = \ln(x + \sqrt{x^2 - 1})$
- Simplify

$$I = x \ln(x + \sqrt{x^2 - 1}) - \sqrt{x^2 - 1} + C$$

Integral #5 (Easy)

$$I = \int_0^1 \frac{x^6}{x^2 + 1} dx$$

Solution Methodology:

- Polynomial Long Division

$$I = \frac{13}{15} - \frac{\pi}{4}$$

Integral #6 (Very Hard)

$$I = \int_0^\infty \frac{x}{e^x - 1} dx \quad \text{Hint: Geometric Series!}$$

Solution Methodology:

- Algebraic manipulation: $\frac{1}{e^x - 1} = \frac{e^{-x}}{1 - e^{-x}}$
- Expand via a geometric series: $\frac{1}{1 - e^{-x}} = \sum_{n=0}^\infty e^{-nx}$
- Simplify Integrand
- Swap summation and integration operators: $\int_0^\infty \sum_{n=0}^\infty = \sum_{n=0}^\infty \int_0^\infty$
- Sub: $t = (n+1)x$
- Separate the integral: $\sum_{n=0}^\infty \int_0^\infty \left(\frac{t}{n+1}\right) e^{-t} \left(\frac{1}{n+1}\right) dt = \sum_{n=0}^\infty \frac{1}{(n+1)^2} \int_0^\infty t e^{-t} dt$
- Identify the Basel Problem and the Gamma Function (or use IBP)

$$I = \frac{\pi^2}{6}$$

Integral #7 (Easy-ish)

$$I = \int_0^{\infty} \frac{\ln x}{x^2 + 1} dx$$

Solution Method:

- Sub: $u = 1/x$
- Simplify
- Observe: $I = -I$

$$\boxed{I = 0}$$

Integral #8 (Easy)

$$I = \int \frac{\pi - e^x}{\pi + e^x} dx$$

Solution Methodology:

- Sub: $u = \pi + e^x$
- Simplify
- Partial Fractions
- Reverse Sub: $x = \ln(u - \pi)$
- Simplify

$$\boxed{I = x - 2 \ln(\pi + e^x) + C}$$

Integral #9 (Medium)

$$\mathcal{L}_t \{ \cos(\omega t - \phi) \} (s) = \int_0^{\infty} e^{-st} \cos(\omega t - \phi) dt$$

Solution Methodology:

- Apply angle sum formula: $\cos(\omega t - \phi) = \cos(\omega t) \cos(\phi) + \sin(\omega t) \sin(\phi)$
- Break up the integral into two integrals:

$$\mathcal{L}_t \{ \cos(\omega t - \phi) \} (s) \cos \phi = \mathcal{L}_t \{ \cos(\omega t) \} (s) \cos \phi + \mathcal{L}_t \{ \sin(\omega t) \} (s) \sin \phi$$

- **Know your Laplace Transforms! Or....**
- Perform IBP $\times 2$ on each integral and recover the implicit recurrence relations
- Solve for the individual integrals
- Combine and simplify

$$\boxed{\mathcal{L}_t \{ \cos(\omega t - \phi) \} (s) = \frac{s \cos \phi + \omega \sin \phi}{s^2 + \omega^2}}$$

Integral #10 (Easy)

$$I = \int \cos(2 \sin^{-1}(x)) dx$$

Solution Methodology:

- Apply Double Angle Formula:
- Simplify

$$I = x - \frac{2x^3}{3} + C$$

Integral #11 (Medium)

$$I = \lim_{h \rightarrow 0^+} \int_0^h \frac{x \cot^{-1}(\pi x)}{\sin(hx)} dx$$

Solution Methodology:

- Expand the sine via its Maclaurin series:

$$\sin(hx) = hx + \mathcal{O}((hx)^3)$$

- Pull out a factor of hx :

$$I = \lim_{h \rightarrow 0^+} \frac{1}{h} \int_0^h \frac{\cot^{-1}(\pi x)}{1 + \mathcal{O}((hx)^2)} dx$$

- Apply the MVT:

$$I = \lim_{h \rightarrow 0^+} \frac{\cot^{-1}(\pi x_h)}{1 + \mathcal{O}((hx_h)^2)}, \quad 0 \leq x_h \leq h$$

- Take the limit via continuity

$$I = \pi/2$$

Integral #12 (Easy)

$$I = \int_{-1}^1 x^2 \cos(\pi x) dx$$

Solution Methodology:

- Even Symmetry (optional)
- IBP

$$I = -\frac{4}{\pi^2}$$

Integral #13 (Medium)

$$I = \int_0^1 x^2 \sqrt{1-x^2} dx$$

Solution Methodology:

- Sub: $x = \sin \theta$
- Simplify Integrand and apply the Double Angle Formula
- Sub: $\phi = 2\theta$
- Apply Double Angle Formula in reverse

$$I = \frac{\pi}{16}$$

Integral #14 (Medium-ish)

$$I = \int_0^{\pi/4} \frac{d\theta}{\cos \theta + \sin \theta}$$

Solution Methodology:

- Apply Angle Sum Formula in reverse: $\cos \theta + \sin \theta = \sqrt{2} \cos(\theta - \pi/4)$
- Sub: $\phi = \theta - \pi/4$, $x = \sin \phi$
- Partial Fractions

$$I = \frac{1}{2\sqrt{2}} \ln \left(\frac{\sqrt{2}+1}{\sqrt{2}-1} \right)$$

Integral #15 (Hard)

$$I = \int_1^\infty \frac{dx}{x^2 \sqrt{x^2 + x - 1}}$$

Solution Methodology:

- Sub: $u = 1/x$
- Simplify
- Complete the square: $1 + u - u^2 = \frac{5}{4} - (u - 1/2)^2$
- Sub: $v = u - 1/2$
- Split integral and abuse odd symmetry
- Sub: $t = \frac{2}{\sqrt{5}}v$

$$\sin^{-1} \left(\frac{1}{\sqrt{5}} \right)$$

Integral #16 (Super Hard)

$$I = \int_0^{\infty} \frac{e^{-t} - e^{-2t}}{t} \cos(t) dt \quad \text{Hint: Laplace Transform}$$

Solution Methodology:

- Define the auxiliary function:

$$F(s) = \int_0^{\infty} e^{-st} \frac{1 - e^{-t}}{t} \cos(t) dt, \quad F(1) = I$$

- Compute the derivative:

$$F'(s) = \int_0^{\infty} \frac{\partial}{\partial s} \left[e^{-st} \frac{1 - e^{-t}}{t} \cos(t) \right] dt = - \int_0^{\infty} e^{-st} (1 - e^{-t}) \cos(t) dt$$

- Split the integral:

$$F'(s) = \int_0^{\infty} e^{-(s+1)t} \cos(t) dt - \int_0^{\infty} e^{-st} \cos(t) dt = \mathcal{L}_t \{ \cos(t) \} (s+1) - \mathcal{L}_t \{ \cos(t) \} (s)$$

- Compute the Laplace Transforms:

$$F'(s) = \frac{s+1}{(s+1)^2 + 1} - \frac{s}{s^2 + 1}$$

- Apply the FTC:

$$F(1) = \lim_{s \rightarrow \infty} F(s) - \int_1^{\infty} F'(s) ds$$

- Compute the limit:

$$\lim_{s \rightarrow \infty} F(s) = \int_0^{\infty} \lim_{s \rightarrow \infty} e^{-st} \frac{1 - e^{-t}}{t} \cos(t) dt = 0$$

- Compute the integral:

$$\begin{aligned} \int_1^{\infty} F'(s) ds &= \int_1^{\infty} \left[\frac{s+1}{(s+1)^2 + 1} - \frac{s}{s^2 + 1} \right] ds \\ &= \left(\frac{1}{2} \ln((s+1)^2 + 1) - \frac{1}{2} \ln(s^2 + 1) \right) \Big|_1^{\infty} = -\frac{1}{2} \ln\left(\frac{5}{2}\right) \end{aligned}$$

- Synthesize

$$I = \frac{1}{2} \ln\left(\frac{5}{2}\right)$$